

Why would anybody hack my PC?

Hackers might break into your computer for various reasons. For one they may want to use your computer as a stepping stone to launch other attacks. This may be to use your machine remotely, to hack into another PC, or to use your PC as part of a distributed denial of service (DDOS) attack. They will use your machine to execute their own programs, which might be anything from password crackers to port scanners. The hackers can also use your hard disk space to host data related to their hacking Web sites, so in case things go wrong, they can easily escape, as a site trace would lead to your computer.

Port scan testing: Port scan testing takes care of the firewall's ability to detect, and resist port scan attempts. Two standard port scanners were used—nmap and LANGuard. A third utility called Port Fun was also used. Finally, Open Ports is an HTA application that loads inside your browser, runs the netstat command locally, and then uploads the results to a server, along with displaying the results in your Web browser.

Detection of existing Trojans: In this test, Port Fun was installed on our test bed prior to installing the firewall. This program opens up ports on the test bed without specifying any service, or application using it. The firewall's ability to block it from opening up the ports was recorded.

OS Detection: Here, nmap was used with the -o option to perform active stack fingerprinting—a technique of sending custom IP packets to the victim's PC, and identifying the OS by analysing the response. This test was performed since detecting the operating system on the victim's computer is one of the first things that a cracker does. It gives him details of the version and patch level of the OS on the target, thereby letting him use relevant exploits available on various underground Web sites.

Hostile scripts test: In this test, the firewall's capability to detect and resist malicious scripts was tested. This has been included because firewall vendors claim that their offerings can detect and protect the host PC from all embedded objects on a Web page. This includes JavaScript, applets, ActiveX objects, etc.

The hostile Java applets used for this test are available at <http://packetstormsecurity.org.pk/java/>. One of these is the Consume Applet whose sole purpose is to waste processor power. To see if the firewall actually inspects all the internals of the HTTP stream, the applet page was renamed with a .txt extension instead of an .html extension.

To test the hostile embedded ActiveX resisting capability of the firewall, we used the ActiveX object test provided at the Finjan.com Web site (<http://www.finjan.com/mcrd/demos/activex.cfm>).

Resource Utilisation

Firewalls protect your network at various security levels—from absolutely no inspection levels for trusted networks, to highly paranoid levels of packet inspection. But all these levels place additional overheads on system resources, proportional to the level of packet inspection specified. All memory resident programs from the system were removed and the firewall was then activated to test resource utilisation. The increase in kernel and memory utilisation was then measured to give the

additional system overhead placed by the firewall in the non-operational state.

Features: Additional features provided by the firewall—Ad Blocking, Mail Scanning, online alerts of malicious activity, update mechanisms, report generation and logging, Traffic Authorisation policies, inbound and outbound connection port blocking, provision for blocking using protocols, security levels provided, and the default security level—were noted.

Test set up: The test set up consisted of three machines. One was the target machine with the firewall installed on it, and the other two were used for hacking into the target. The target computer ran Windows XP. One of the attacking machines also ran XP, while the other was Linux based. The Linux machine used nmap for all the scanning and OS fingerprinting tests, while the Windows PC used LANGuard.

Sygate Personal Firewall Pro v5.0 build 1175:

This firewall is pleasing to the eye, and instead of a list of hacking events that your PC is subject to, it displays eye-catching graphs that show all the traffic in your little corner of the Internet and the work done by villains trying to get into your system. Also, it logs almost all the Internet activity into four logs—the Security Log, Traffic Log, Packet Log and System Log. It also gives you a list of all the applications currently accessing the Internet in their full iconic glory. It also does a very good job at stopping port scans, going as far as specifically stating that nmap was trying to do a port scan on your machine. It was also one of the few firewalls that stopped Open Ports from loading directly from the Web site. However it was toothless when Open Ports was run locally; it could not stop it from uploading

Info

The CyberArmy is a shady group of hackers that perform Distributed Denial Of Service (DDOS) attacks, in a claim to 'liberate' the Internet from authorities

